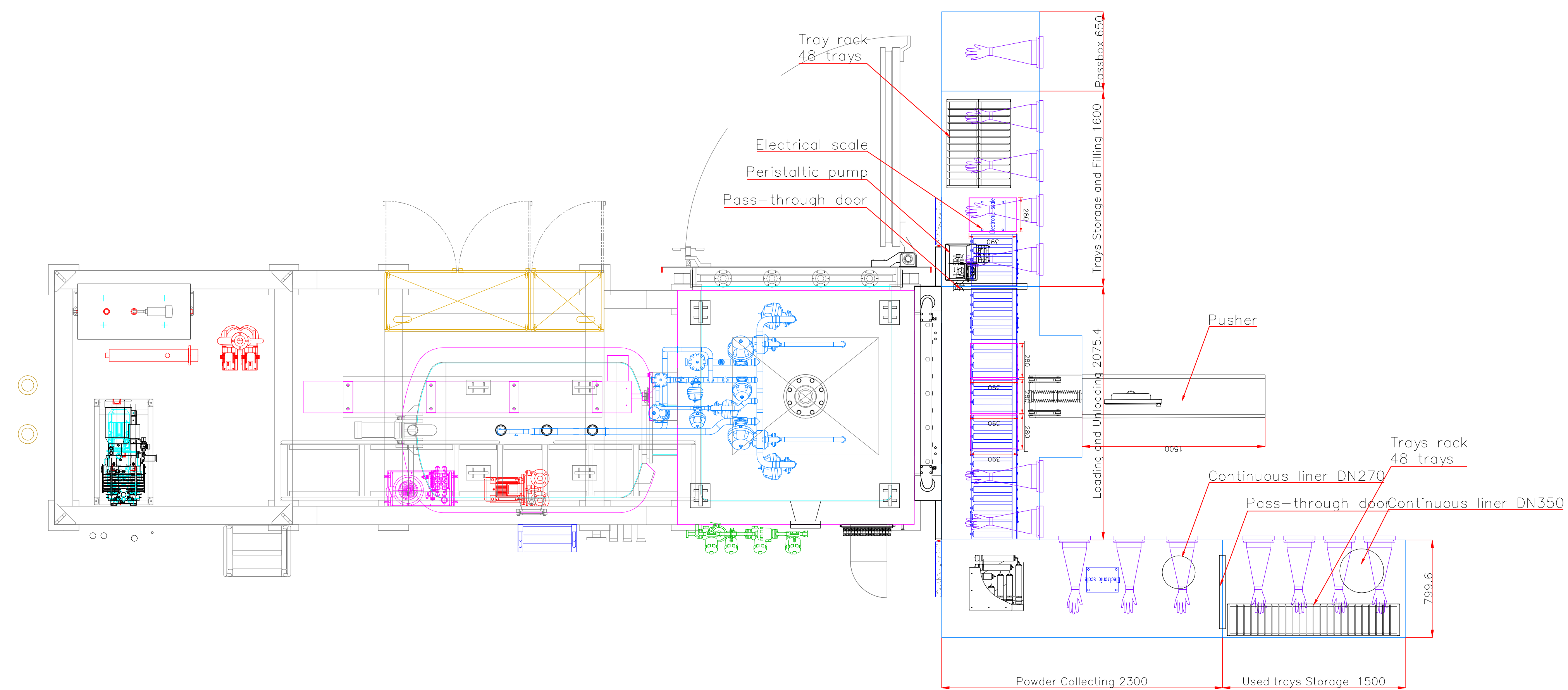


方案2： 一次性托盘

Solution 2: Single use Trays



- 工艺描述：
- 1、加料枪与蠕动泵相连；客户物料罐根据物料罐形式直接用软管接入或通过硬管连接；
 - 2、蠕动泵为单泵头，与加料枪相连，人工手持蠕动泵软管对托盘定量加料，蠕动泵设定好一盘加料量（加料枪需要固定，托盘下方设置电子秤~量程6kg，精度1g）；
 - 3、冻干机板层先降到最低，小车从上到下依次将托盘推至冻干机板层上；
 - 4、三个托盘一组进出冻干箱板层上，一块板层9个托盘，进出三次；单组加料结束，推杆将加料托盘送入等高位板层，依次重复前面操作，直到一块板层铺盘结束；
 - 5、冻干机由上至下依次进行铺盘；所有板层铺盘结束后关闭冻干小门进行冻干；
 - 6、生产过程中，负压隔离器工作，满足OEB5泄露要求后，转轮除湿机给隔离器提供新风，湿度控制在≤15%；
 - 7、冻干后，冻干小门自动打开，后推杆将托盘逐层推出到滚筒上。
 - 8、托盘转移至收料段，人工手动收粉至袋中；
 - 9、循环步骤7—8，直至所有物料装入袋中；
 - 10、收料完毕后，分装袋通过DN270连续袋出装置传出；
 - 11、托盘传入冻干盘预处理舱段，每两个托盘一组进行套袋，托盘通过DN350连续袋出装置传出。

- Process description:
1. The feeding gun is connected to the peristaltic pump; the customer's material tank is directly connected to or connected through a hard pipe according to the form of the tank.
 2. The peristaltic pump has a single pump head and is connected to the feeding gun. The operator holds the peristaltic pump hose manually to add the material to the tray in a quantitative manner. The peristaltic pump is set to a certain amount for one tray (the feeding gun needs to be fixed, and an electronic scale is set below the tray – the measurement range is 6kg, and the accuracy is 1g).
 3. The shelves of the freeze dryer are lowered to the lowest position. The pusher pushes the trays to the shelves from bottom to top.
 4. Three trays form a group, pusher pushes three trays one time. One shelf has 9 trays, and this process is repeated three times. After the filling is completed for one group of trays, the pusher moves the feeding tray into a shelf, and the above operations are repeated until all shelves are fully loaded with trays.
 5. The freeze dryer shelves are loaded with trays from top to bottom in sequence. After all shelves are loaded with trays, the freeze dryer door is closed for freeze drying.
 6. During the production process, the negative pressure isolator operates. After meeting the OEB5 leakage requirements, the rotary dehumidifier provides fresh air to the isolator, and the humidity is controlled at ≤15%.
 7. After freeze drying, the pizza door automatically opens, and the back pusher pushes the trays shelf by shelf out to the roller.
 8. The trays are transferred to the receiving section, and the powder is manually collected into bags by the operator.
 9. Steps 7–8 are repeated until all materials are loaded into bags.
 10. After the receiving is completed, the packaging bags are transmitted through the DN270 continuous bag–out device.
 11. The trays are transferred to the freeze dryer plate pre–treatment section. trays are grouped into pairs for bagging, and then transferred through the DN350 continuous bag–out device.